



ARC RESEARCH · WEEKLY AI UPDATE · SEPT 19–25, 2026

Cheaper frontier models, efficient open weights, and China's chip push

As model performance converges, labs race to build their own chips or go deep tech

Weekly briefing · Data as of Sept 25, 2026 · Vendor-reported numbers are labeled · Educational purposes only, not investment advice

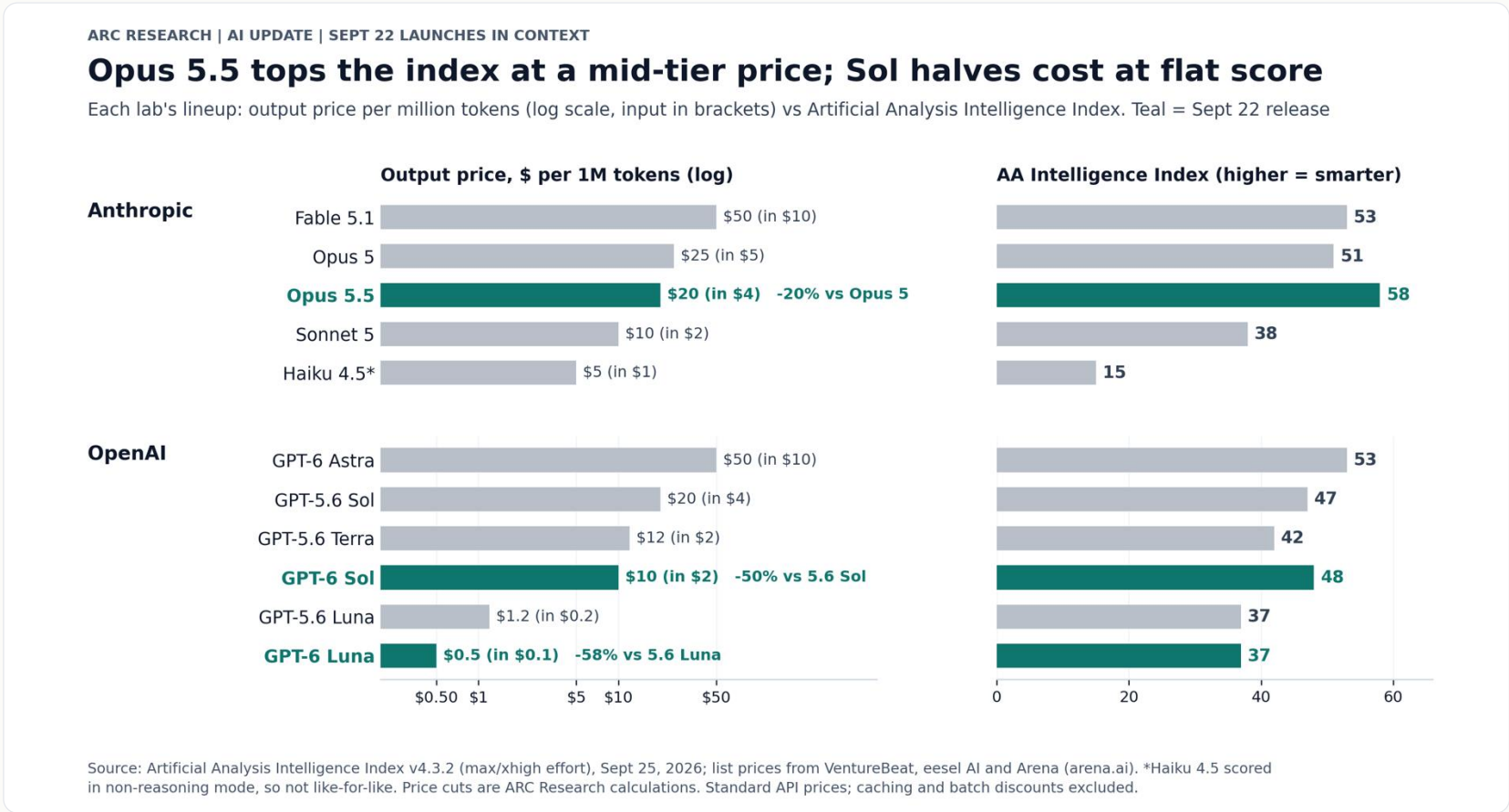
THIS WEEK

Four stories, one theme: more intelligence per dollar

01	Closed models	Claude Opus 5.5 vs GPT-6 Sol/Luna: benchmarks, cost per task, strategy
02	Open weights	Xiaomi MiMo tops open models; rumored releases fact-checked
03	Qwen at Apsara	Qwen 4 in training, self-improving Qwen3.8-Max, Qwen Intelligence
04	AI chips	Alibaba Zhenwu V900 and Huawei's faster Ascend roadmap



Shipped 90 minutes apart, both cheaper



SO WHAT

Opus 5.5 scores highest (58) yet costs 60% less than Fable 5.1 (53)

GPT-6 Sol: +1 point for half the price; Luna: same score, 58% cheaper

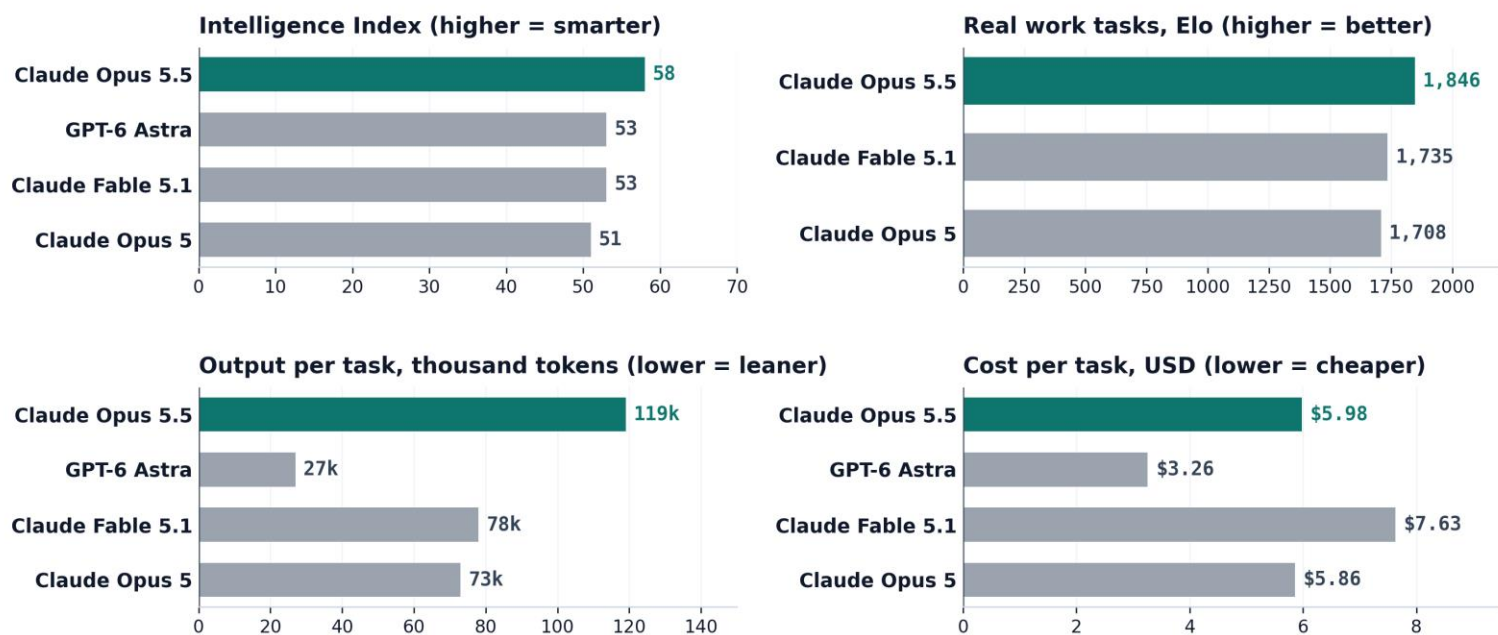
Both flagships (Fable 5.1, Astra) score 53 at \$10 / \$50

Opus 5.5: smartest model tested, and the wordiest

ARC RESEARCH | AI UPDATE | CLAUDE OPUS 5.5 PERFORMANCE

Opus 5.5 is the smartest model tested, and the wordiest

It leads on intelligence and real-work tasks, but writes far more per task, so cost per task stayed flat



Source: Artificial Analysis (AA) Intelligence Index, output tokens and cost per Index task, max effort (Sept 22-25, 2026). Work-task Elo: GDPval-AA v2.1, Opus 5.5 from AA; Fable 5.1 and Opus 5 from Anthropic's launch table via AlphaCorp. Higher Elo = wins more head-to-head comparisons on real job tasks.

SO WHAT

#1 on the independent index, 5 points clear

Best at real job tasks in head-to-head tests

20% cheaper per token, but it writes more, so a task costs the same

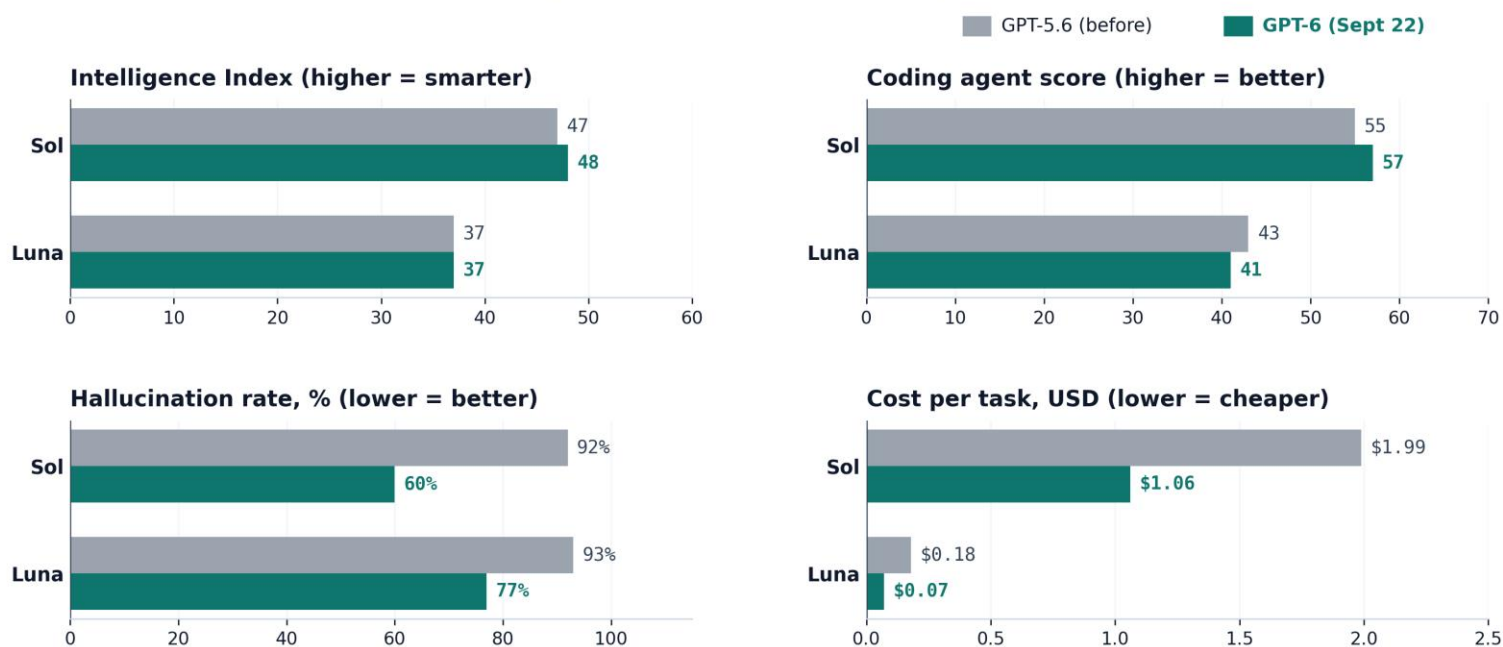


Sol and Luna: about as smart, far cheaper

ARC RESEARCH | AI UPDATE | GPT-6 SOL AND LUNA PERFORMANCE

GPT-6 Sol and Luna: about as smart as before, far cheaper

Scores barely moved and hallucinations fell sharply, while the cost to finish a task roughly halved



Source: Artificial Analysis Intelligence Index, Coding Agent Index, AA-Omniscience hallucination rate and cost per Index task, max effort, via AA and OfficeChai (Sept 22-23, 2026). Grey = GPT-5.6 predecessor, teal = GPT-6. Hallucination rate = share of answers that were confidently wrong instead of declining.

SO WHAT

Intelligence scores barely moved

Far fewer confidently wrong answers

A task now costs about half as much, or less



Anthropic bought intelligence, OpenAI bought efficiency

ARC RESEARCH | AI UPDATE | INTELLIGENCE VS COST PER TASK

Anthropic bought intelligence; OpenAI bought efficiency

Opus 5.5 gained 7 index points at a flat cost per task; Sol and Luna held their scores at about half the cost



Source: Artificial Analysis Intelligence Index and cost per Index task, max effort, from AA model pages and launch articles (Sept 22-25, 2026), incl. MiMo-V2.6-Pro. Independent evaluation. Cost per task = AA's measured spend to run one Index task at list prices, so it reflects both price and tokens used. Arrows: previous model -> new model. Log x-axis.

SO WHAT

Opus 5.5: +7 points, cost per task flat (+2%)

Sol and Luna: same score, cost per task -47% and -61%

Open MiMo-V2.6-Pro: 46 at 1/46th of Opus's cost

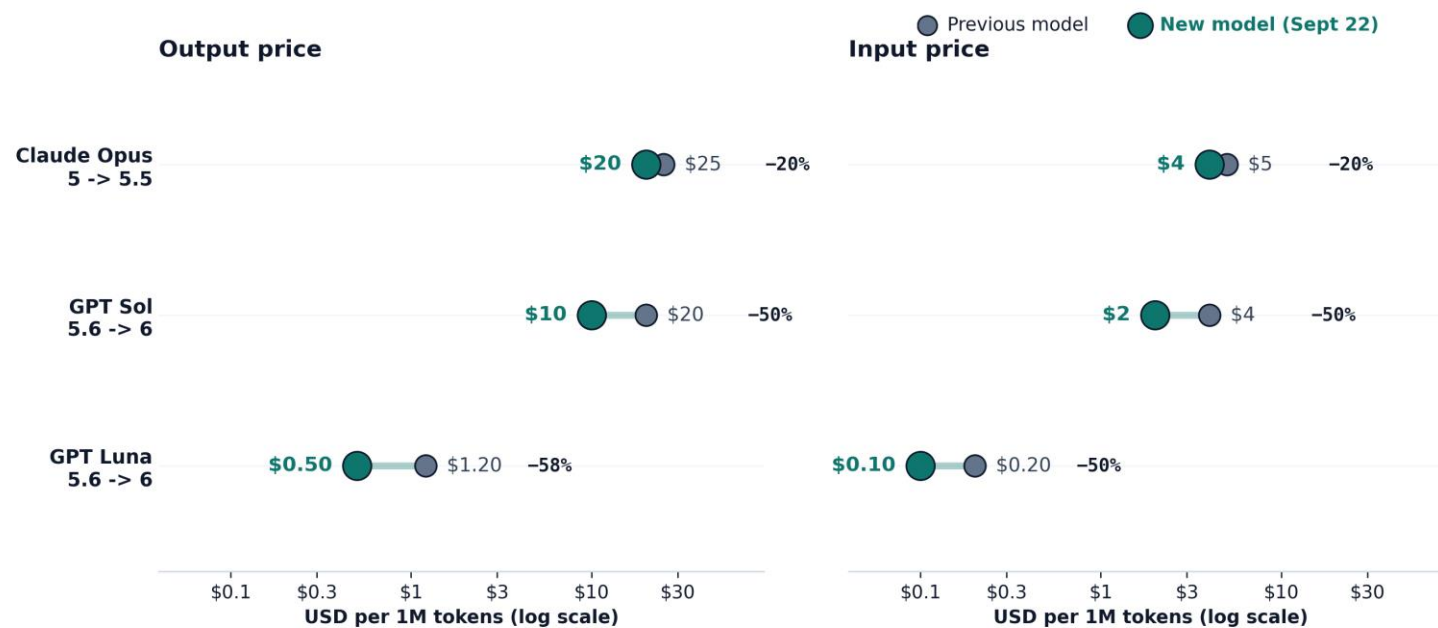


Output prices dropped 20–58% in a single day

ARC RESEARCH | AI UPDATE | MODEL PRICING, SEPT 22 2026

Opus 5.5 and GPT-6 Sol/Luna cut API prices 20-58% on one day

Luna now costs about 1/40th of Opus 5.5 per token, pushing developers to route by task



Source: Quartz (Opus 5 and 5.5 list prices); The New Stack and VentureBeat (GPT-5.6 and GPT-6 list prices), Sept 22-23, 2026.
Reported list prices, USD per 1M tokens, uncached, standard tier. % change = new / old - 1 (ARC Research calculation). Log scale.

SO WHAT

OpenAI halved Sol; Luna output fell 58%

Opus 5.5 is 20% cheaper; cache reads fell 60%

Luna costs ~1/40th of Opus 5.5 per token



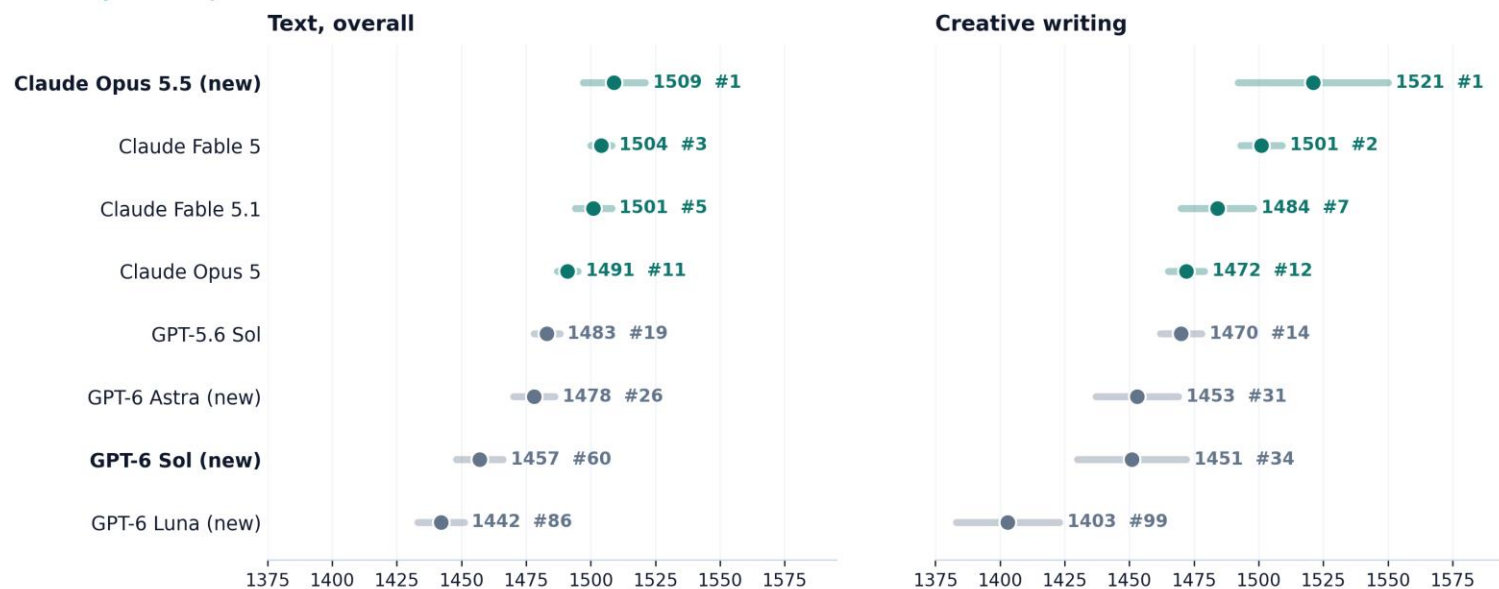
Real users prefer Opus 5.5; GPT-6 Sol slipped

ARC RESEARCH | AI UPDATE | WHAT REAL USERS PREFER

In blind votes, people pick Opus 5.5 first; GPT-6 Sol slipped

Arena score by model, overall text vs creative writing (higher = preferred more often in head-to-head votes)

● Anthropic ● OpenAI



Source: Arena (arena.ai) Text leaderboard, snapshot Sept 25, 2026 (8.5M votes). Score = Elo-style rating from blind head-to-head votes; whiskers = Arena's 95% interval. Opus 5.5 has few votes so far (2,307 overall; 487 creative), so its interval is wide and its rank is preliminary. Anthropic = teal, OpenAI = slate.

SO WHAT

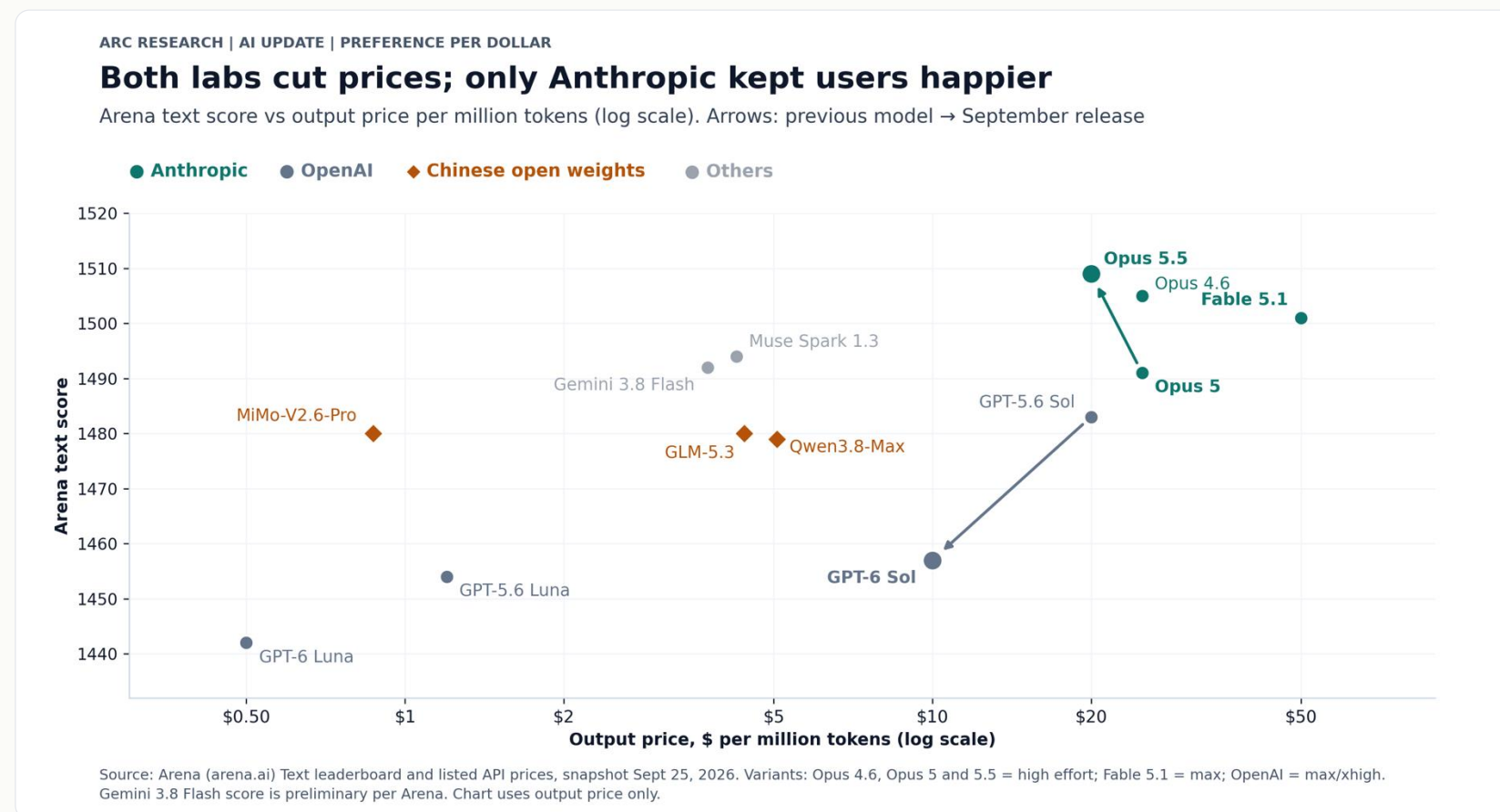
Opus 5.5 is #1 overall, in creative writing and in instruction following

Anthropic targeted its writing: fewer walls of text, key point first

GPT-6 Sol ranks below the model it replaced (#60 vs #19)



Both cut prices; only Anthropic won users



SO WHAT

Both labs are now selling intelligence per dollar, not just top scores

Opus 5.5: 20% cheaper than Opus 5 and preferred more

Sol: half the price, but users like it less than GPT-5.6 Sol



Why Opus 5 and Fable 5.1 disappointed

1491 < 1505

Opus 5 scored below older Opus 4.6 on Arena

Users said it “never answers the actual question” (MindStudio)

11%

Fable 5’s share of Anthropic’s enterprise revenue

Ramp data via FT; at \$10/\$50 per M tokens, firms chose cheaper models

2.5×

Fable 5.1’s price vs Opus 5.5

\$50 vs \$20 per M output tokens, yet ranked lower on Arena (1501 vs 1509)

Benchmarks vs real use: Opus 5 matched GPT-5.6 on coding and agent tests, but many users rated it worse than Opus 4.8 in daily work.

Not a total miss: reviewers called Fable 5.1 a real capability jump. Its problem was price, not ability.

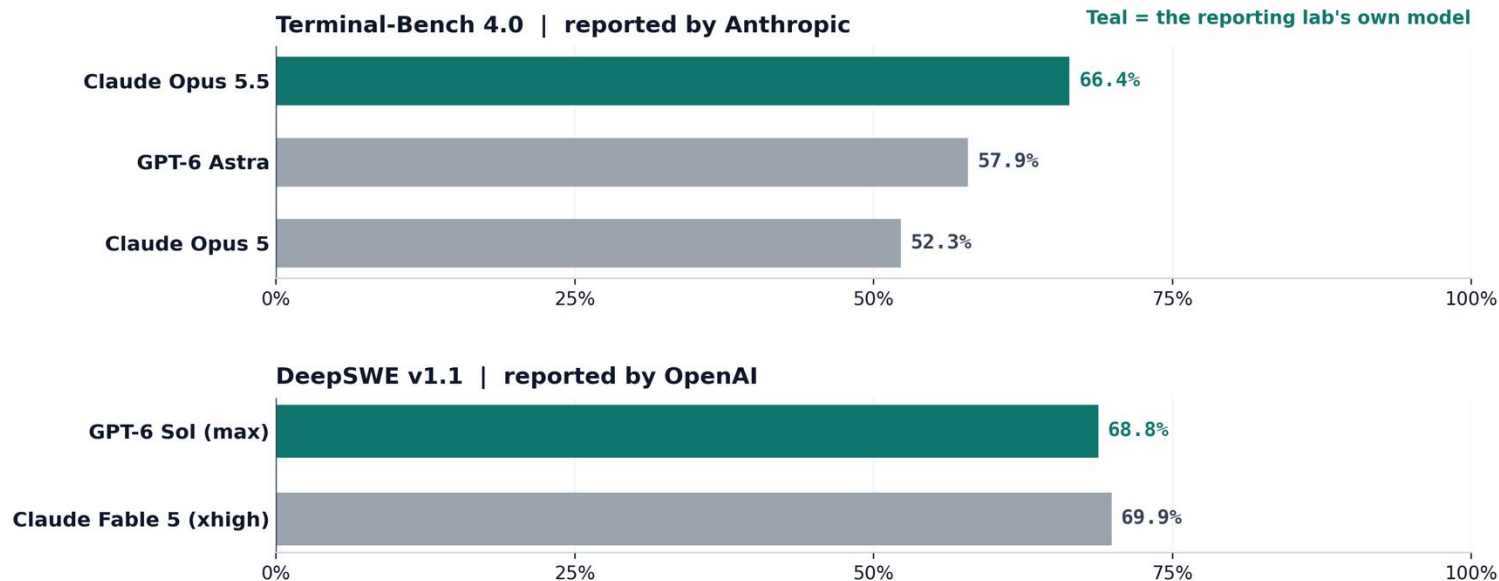


Each lab's launch benchmark flatters its own model

ARC RESEARCH | AI UPDATE | VENDOR-REPORTED CODING BENCHMARKS

Each lab's launch benchmark flatters its own new model

Anthropic's table puts Opus 5.5 8.5 pts ahead; OpenAI's shows Sol near Fable 5 at about 20% of the cost



Source: Anthropic launch benchmark table via AlphaCorp (Terminal-Bench 4.0); OpenAI results via The New Stack (DeepSWE v1.1), Sept 22-23, 2026.
All scores vendor-reported, % of tasks solved; effort settings as published (Sol at max, Fable 5 at xhigh). No independent evaluation at launch.

SO WHAT

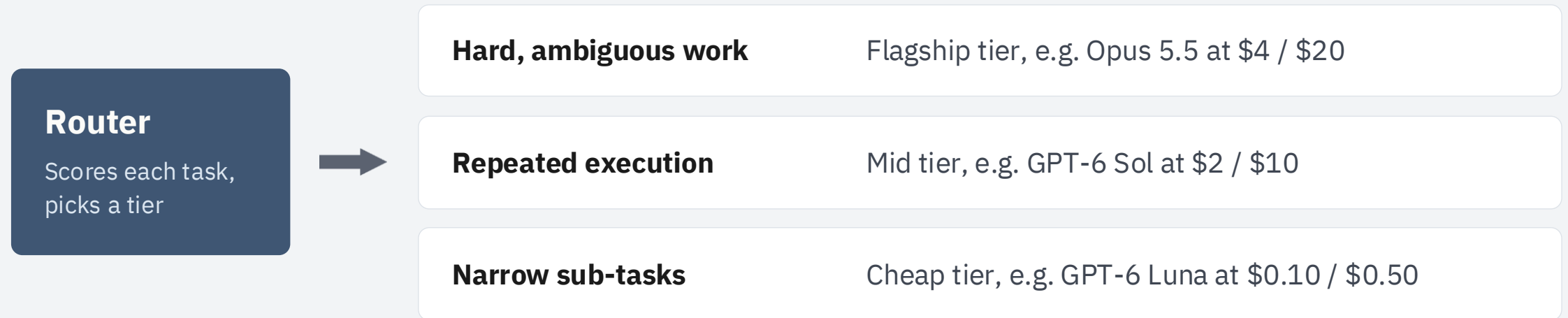
Anthropic's table: Opus 5.5 leads by 8.5 pts

OpenAI's table: Sol near Fable at ~20% of the cost

All company-reported; ask who ran the test



Price tiers push developers toward model routing



Squeezed on coding, labs bet on bio and robots

THE SQUEEZE · CODING AND APIS

23× cheaper output: Xiaomi's open MiMo-V2.6-Pro (\$0.87/M) vs Opus 5.5 (\$20/M), about 30 Arena points apart

-41% average price businesses pay per token since March (Ramp)

11% of Anthropic's enterprise revenue from its priciest model, Fable 5 (Ramp via FT)



THE HEDGE · HARDER-TO-COPY BETS, 2026

APR 3 **Anthropic** buys drug-discovery startup Coefficient Bio (~\$400M)

MAY 12 **Google** Isomorphic Labs raises \$2.1B to design drugs with AI

MAY 31 **OpenAI** launches a robotics division; 19 open roles by Sept

JUN 30 **Anthropic** launches Claude Science for biology and drug research

SEP 3 **OpenAI** Altman: "We will definitely do a humanoid"

Reality check: a hedge, not an exit. Only 3.6% of businesses use open models (Ramp), and coding revenue is still growing fast.

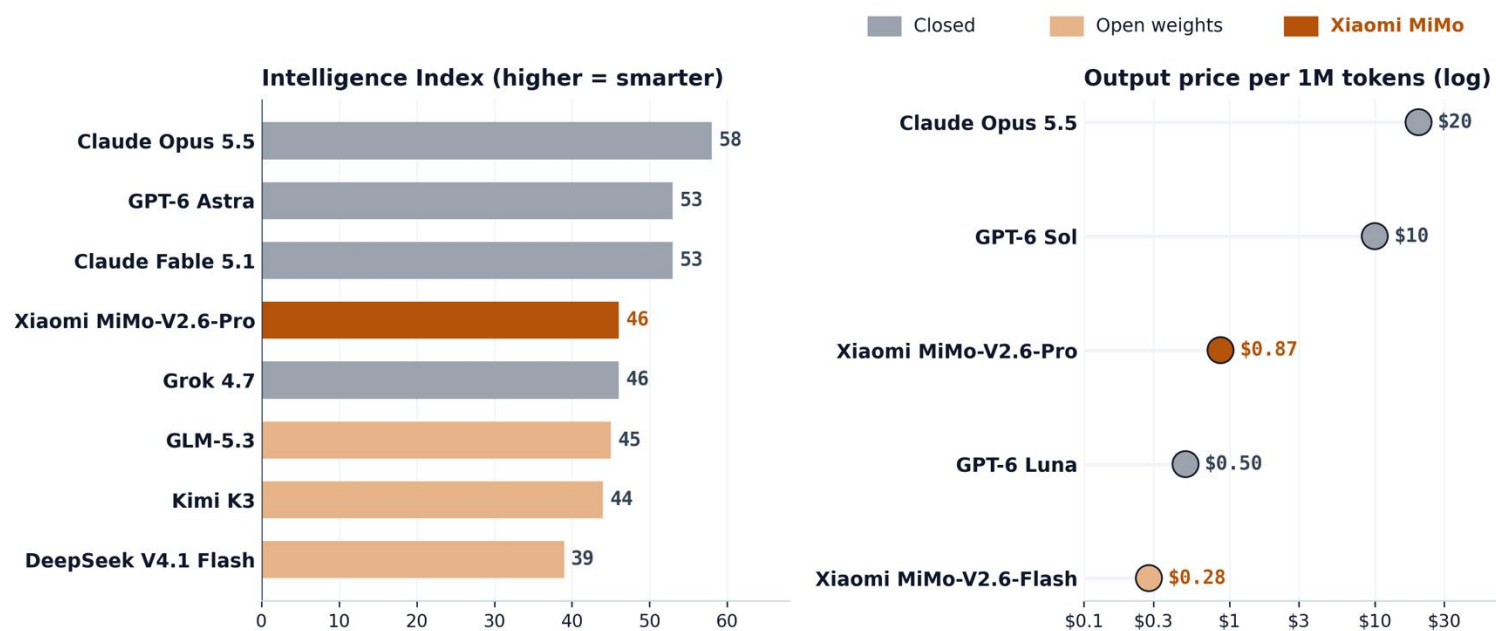


Xiaomi's open MiMo trails leaders at a tiny price

ARC RESEARCH | AI UPDATE | OPEN VS CLOSED MODELS

Xiaomi's free-to-download MiMo trails the leaders at a fraction of the price

MiMo-V2.6-Pro is the top open-weights model on the independent index, 12 points behind Opus 5.5



Source: Artificial Analysis Intelligence Index via OfficeChai (Opus 5.5, Astra, Fable 5.1, Grok 4.7, MiMo), TechNode (GLM-5.3, Kimi K3), VentureBeat (DeepSeek V4.1 Flash), Sept 22-25, 2026. Prices: list price per 1M output tokens (Xiaomi, OpenAI via VentureBeat; Anthropic via Quartz).

SO WHAT

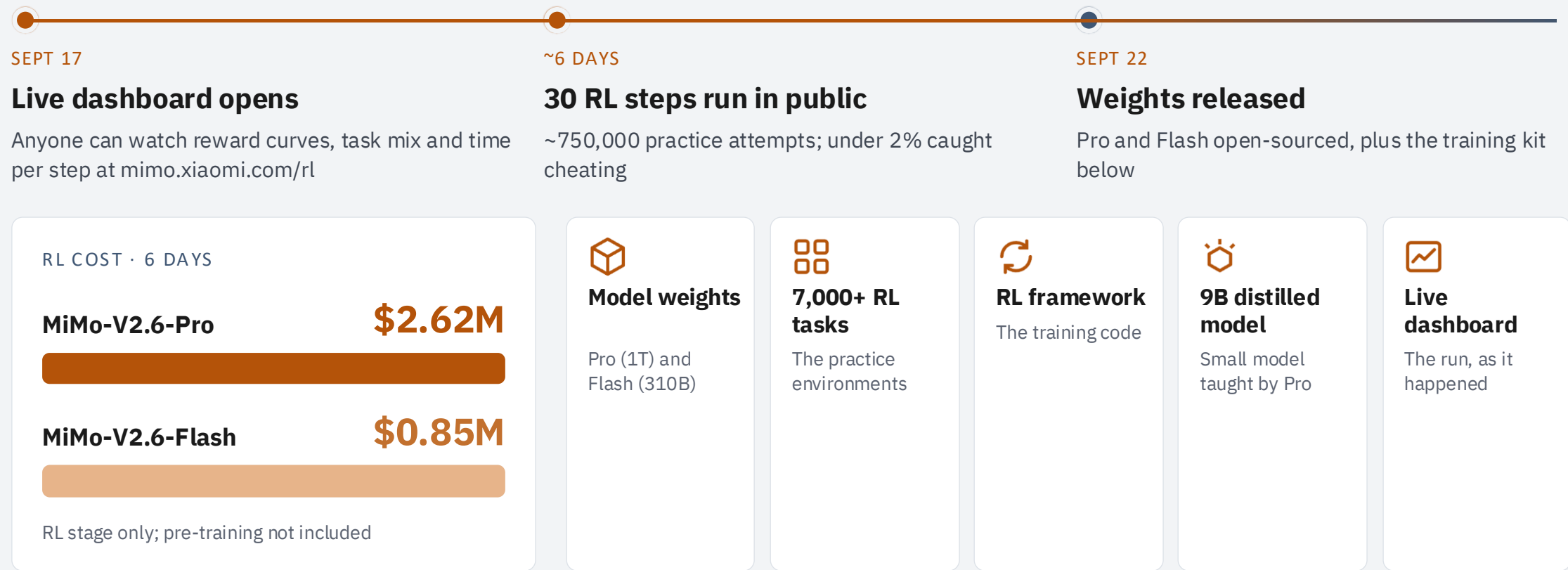
Best free-to-download model, level with Grok 4.7

Still 12 points behind Opus 5.5

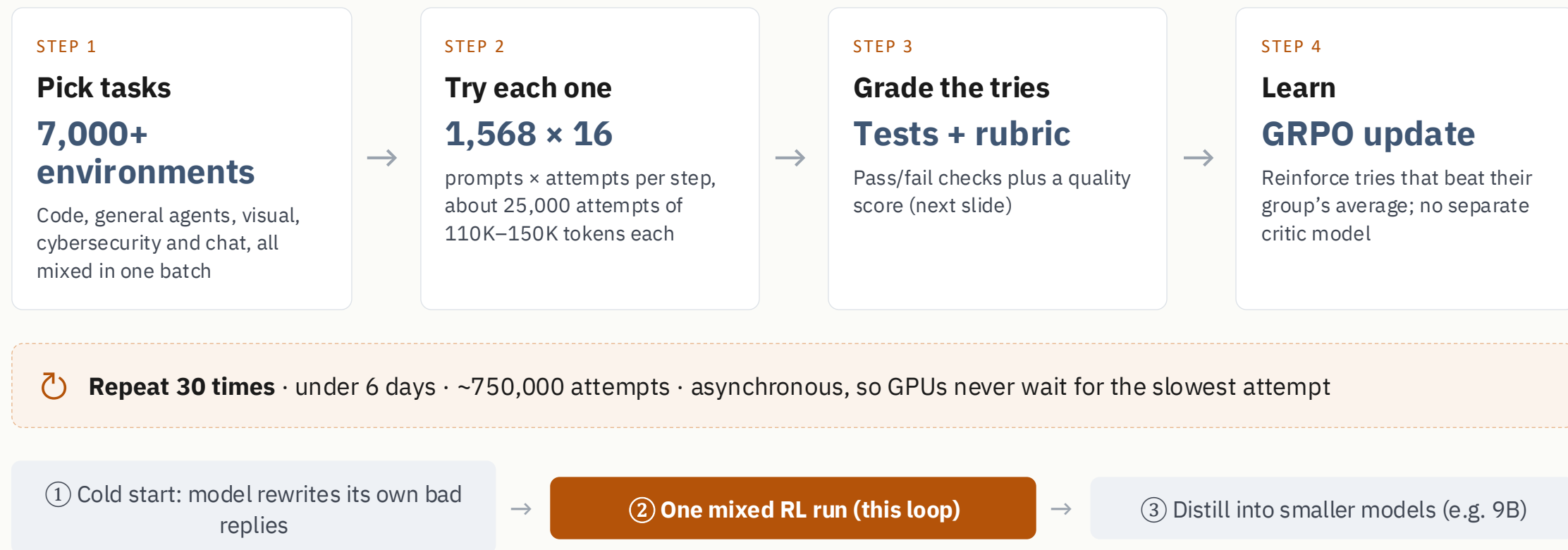
Output costs \$0.87 per 1M tokens vs Opus's \$20



Trained in public, then open-sourced the kit



One training run taught every skill at once



Rewarding good work, not just passing work

THE PROBLEM

Two fixes can both pass the tests while one is messy and ignores the codebase. Pass/fail can't tell them apart.

GRS

Compares the 16 tries to write a grading rubric for that task (edge cases, respecting the code, gathering evidence), then mixes it with the test result

GAR

Ranks the tries that passed and shifts the learning signal toward the best ones

Effect: without it, step counts and output length ballooned; with it, they stayed stable

BLOCKING “REWARD HACKING”

Removed build files and caches the model could copy

Deleted “future” Git history holding the answers

Cut internet access to answer sources

Ran a separate “hack agent” hunting for loopholes

<2% of attempts confirmed as cheating

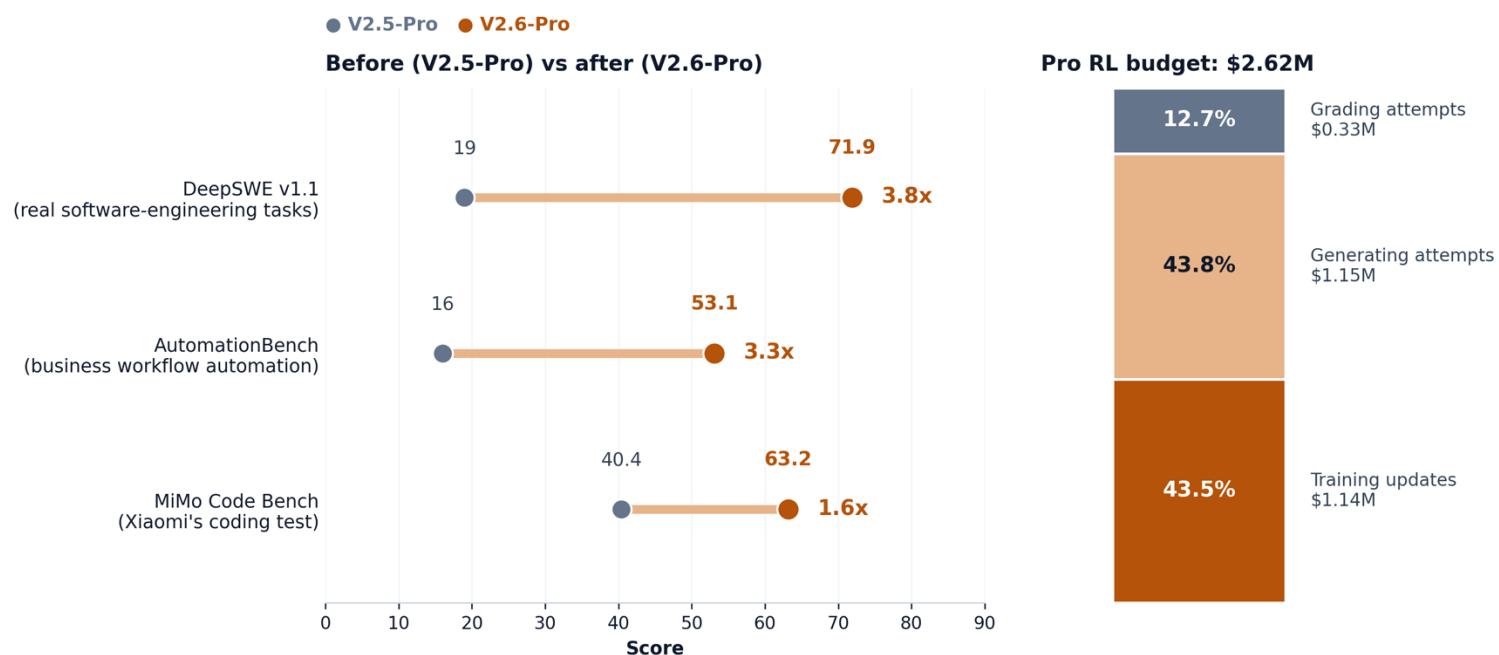


Agent scores up to 3.8x, for a \$2.62M run

ARC RESEARCH | AI UPDATE | XIAOMI MIMO-V2.6 POST-TRAINING

Agent scores jumped up to 3.8x; practice runs cost as much as learning

MiMo-V2.6-Pro vs MiMo-V2.5-Pro on agent and coding tests, and where the \$2.62M RL budget went



Source: Xiaomi via VentureBeat and the MiMo-V2.6-Pro-RL model card (Sept 2026). Scores are Xiaomi-reported; MiMo Code Bench is Xiaomi's own test. Gains reflect the whole V2.6 release, which Xiaomi does not split between RL and other training stages. RL cost excludes pre-training. Multiples are ARC Research calculations.

SO WHAT

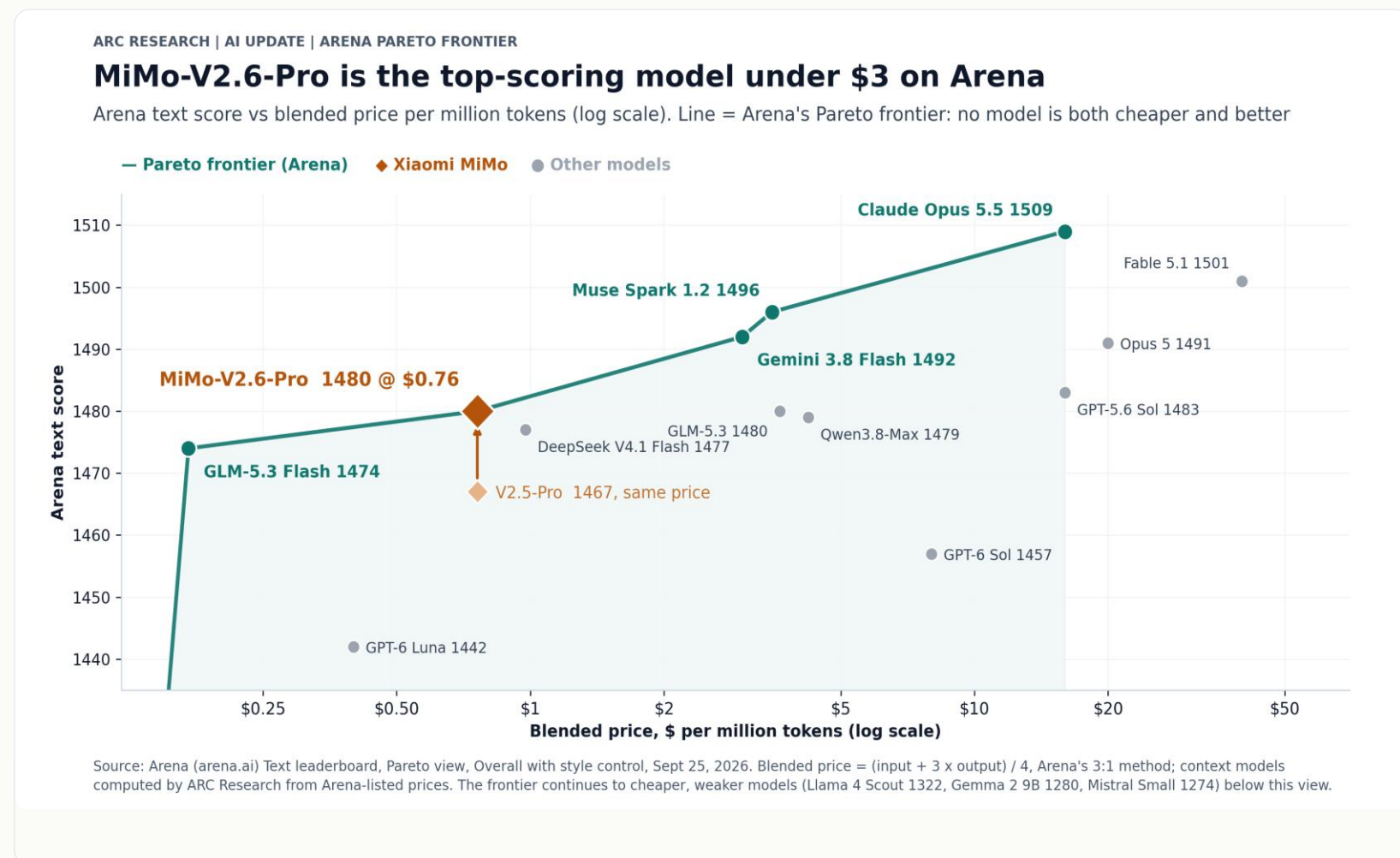
Top open-weights model:
Artificial Analysis index 46, tied
with Grok 4.7

Generating practice attempts
cost as much as learning from
them

Still behind Opus 5 on
DeepSWE and GPT-5.6 Sol on
hacking tests



MiMo made Arena's best-value frontier



SO WHAT

On the frontier: nothing cheaper scores higher

+13 points over V2.5-Pro at the same \$0.76 price

29 points behind Opus 5.5 at 1/21 of the price



Qwen 4 is in training, and Qwen3.8-Max improved itself

The roadmap

Qwen 4: in training, no date, specs or price yet

Qwen 4.5 / Qwen 5: projected 5–10 trillion parameters

Today's flagship Qwen3.8-Max: 2.4 trillion

Recursive self-improvement

33 automated cycles over 1+ month

Artificial Analysis score rose from 40 to 45

Chip-design test: 10,000+ EDA tool calls, 42% smaller chip area

Preview of Qwen 4's architecture: Qwen3.8-Flash-Next, open weights since Aug 26.



Qwen Intelligence is a full agent stack sold to phone makers

TOP	Ready-made agents Planner · Mobile-Use · Creative	Launched Sept 22 at Apsara, Hangzhou
MIDDLE	Customizable platform Model deployment, tool governance, auto-evaluation, device-cloud split	Customer: phone makers, not consumers (B2B)
BASE	Mobile-optimized Qwen models Smaller models that can run on the phone itself	Pricing: not disclosed

Three agents split planning, doing and creating

Mobile Planner

Breaks a request into steps, picks tools, adjusts as it goes

e.g. a business trip: flights, hotel, calendar, itinerary

Mobile-Use

Acts across apps: calls an app's API first; if none, taps the screen

"API-first, GUI-fallback"

Mobile Creative

Lightweight model for image generation and editing

100+

steps in one workflow

≤91.8%

task accuracy (benchmark not named)

1,000+

MobilePA-Bench tasks, 200+ tools



HONOR ships it first, on the Magic9 from Sept 28

Sept 22	Alibaba launches Qwen Intelligence at Apsara
Sept 28	HONOR Magic9 series global launch, first phones with it
TBA	HONOR Robot Phone, also named as a first device

Why it matters

Qwen reaches users through other brands' phones

Alibaba calls agentic phones the main gateway from personal AI to the physical world

Open question: privacy when an agent taps through your apps

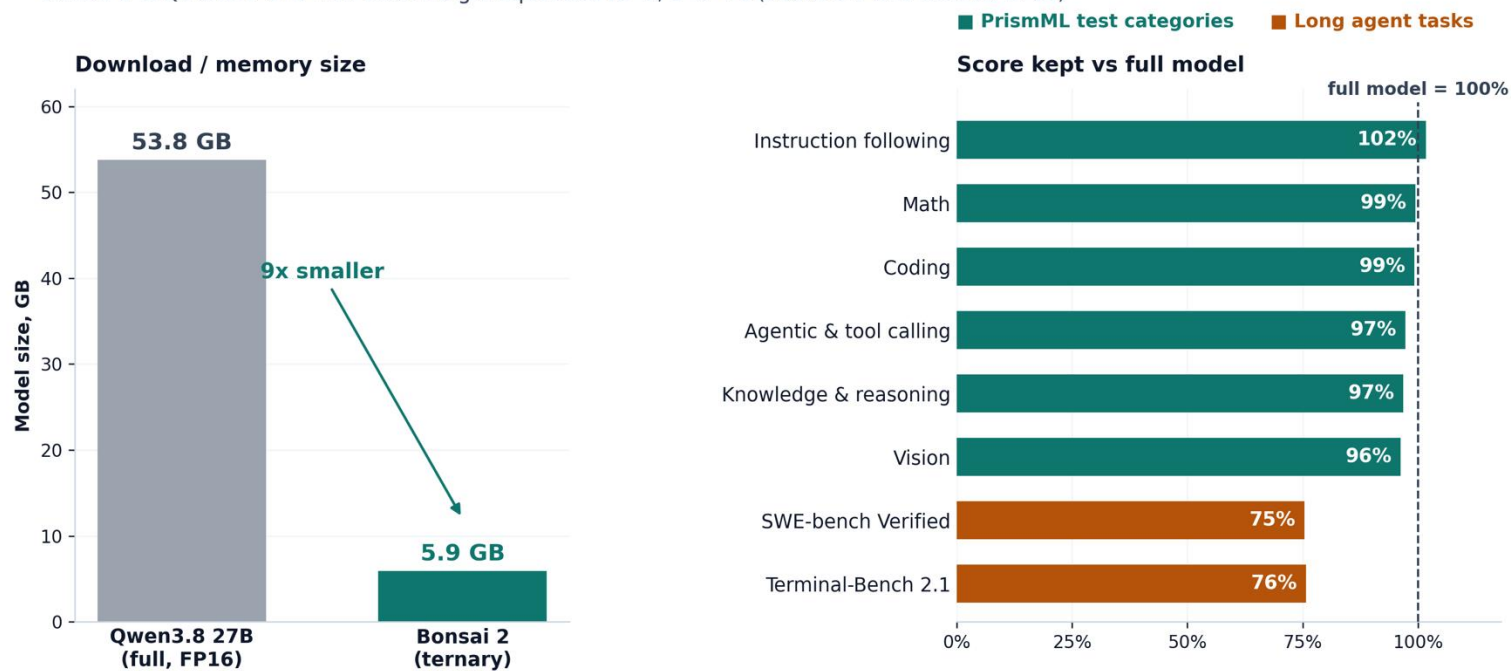
Also at Apsara: Qwen3.8-LiveTranslate (latency 2.8s → 2.3s), Qwen-Audio 3.1 suite, Qwen-Image 3.1 later in 2026.

Bonsai 2 shrinks a 27B model 9x, keeps 98%

ARC RESEARCH | AI UPDATE | PRISMML TERNARY BONSAI 2 27B

9x smaller, near-full scores on short tasks, weaker on long agent work

Bonsai 2 is Qwen3.8 27B with each weight squeezed to -1, 0 or +1 (about 1.7 bits instead of 16)



Source: PrismML announcement (Sept 17, 2026) and MarkTechPost (Sept 18, 2026). All scores are PrismML's own and not independently reproduced.
Retention = Bonsai 2 score / full-precision Qwen3.8 27B score (ARC Research calculation for SWE-bench Verified and Terminal-Bench 2.1).

SO WHAT

5.9 GB: a Qwen-class model on one gaming GPU or a Mac, free

143 tokens/sec on an RTX 5090; 47 on an M5 Max Mac

Weak spot: long coding-agent tasks keep only ~75%



Five rumored models, and how solid each rumor is

Model	What exists	Best evidence	Timing signal	Evidence
DeepSeek V4.1 Pro	Named by DeepSeek; OpenCode placeholder	DeepSeek's Sept 10 note: old traffic moves "until V4.1-Pro launches"	No date; maybe before Oct 1	Strong
Muse Spark 1.4 (Meta)	OpenCode placeholder, 2 test sessions	Betting markets; The Information: Meta's next model targets October	84% by Oct 16 (Polymarket)	Medium
Kimi K4 (Moonshot)	OpenCode placeholder, no use	The Information: Moonshot seeks more Nvidia chips for a bigger model	22% in 2026 (Manifold)	Medium
GLM-5.4 (Z.ai)	OpenCode placeholder, no use	Mystery model "Omen Alpha" linked to Z.ai by testers, unconfirmed	Cadence: Oct 8–Nov 9	Weak
GLM-5.5 Flash (Z.ai)	OpenCode placeholder only	No leak; a larger GLM-5.5 expected in August never shipped	None	Weak

Placeholder = OpenCode lists the name but has recorded no real use. None has confirmed specs, price or date.

The "DeepSeek V5" leak: half reported, half hype

Claim in the viral post	Verdict	Source
A 2-trillion-parameter model is in training	Reported	The Information, Sept 22
Founder calls it one of "our biggest bets"	Reported	The Information (unnamed)
Needs ~4x more Huawei chips than Nvidia	Reported	Founder's estimate, via The Information
It is called "V5"	Unverified	No report uses the name
Trained fully on Huawei Ascend chips	Contradicted	Bloomberg: training stays on Nvidia
Beats OpenAI's GPT-6 Astra	No evidence	No benchmark exists

Chips to train a top model

~200,000

Huawei Ascend 950

~50,000

Nvidia GB300 (founder's estimate)

160,000

Huawei chips ordered for a DeepSeek data center, to run models, not train them

Alibaba's Zhenwu V900 claims 3× its predecessor

Spec	Zhenwu V900 (Sept 22)	Zhenwu M890 (May)
Memory per chip	216 GB	144 GB
Chip-to-chip bandwidth	1,200 GB/s	800 GB/s
Precision	FP32 down to FP8 and FP4	—
Max cluster	500,000 chips	—
Absolute FLOPS · foundry	not disclosed	—
Availability	Mass production Q1 2027	Shipping

The "3×" is vendor-reported and can't be compared with Nvidia without FLOPS.



Huawei pulled its next Ascend chip forward to early 2027

Q1 2027	Q3 2027	2028	2029
Ascend 960DT	Ascend 960PR	Ascend 970	Ascend 980
Moved earlier from Q3 2027	192 GB, 2.4 TB/s memory bandwidth	~2× compute per generation	Per Huawei's roadmap

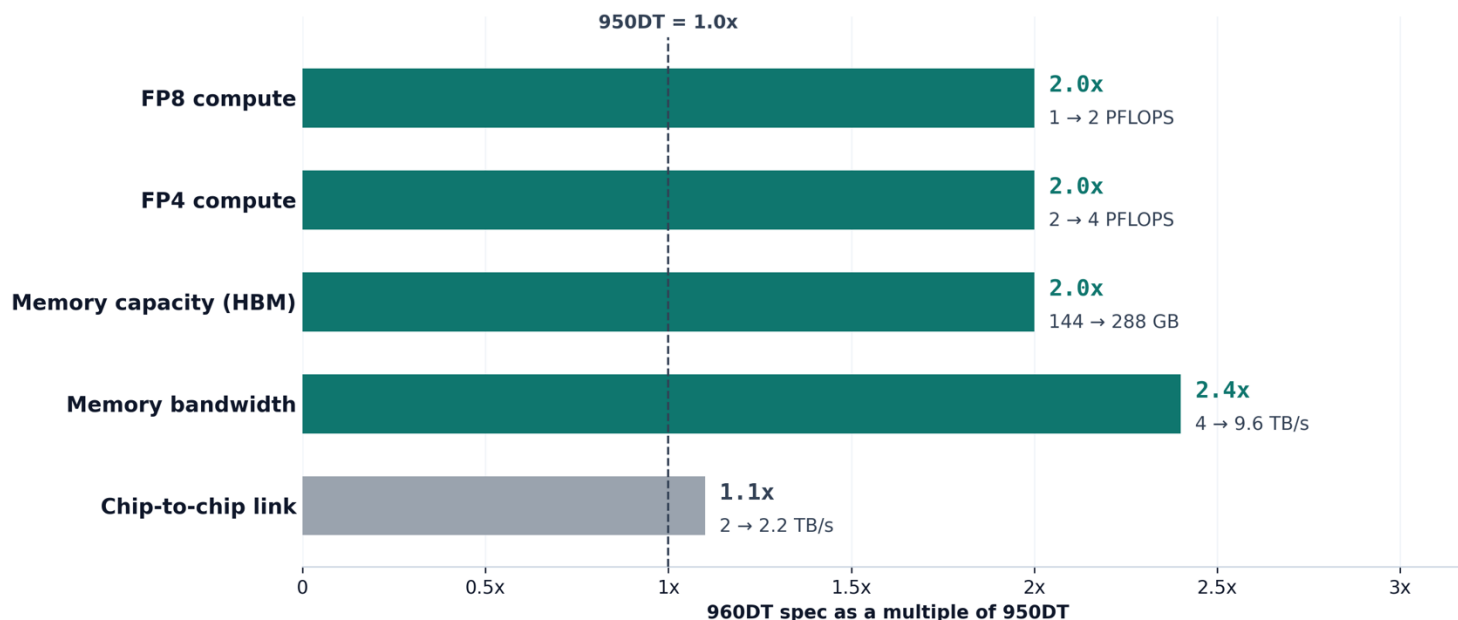
Per chip, Huawei still trails Nvidia's best (WSJ); it compensates with clusters of up to 256,000 accelerators.

What Huawei is upgrading in its next chip

ARC RESEARCH | AI UPDATE | HUAWEI ASCEND 950DT → 960DT

Huawei's next chip doubles compute and memory, not the chip link

Ascend 960DT (due Q1 2027) vs 950DT (shipping 2026): the biggest jump is memory bandwidth, 2.4x



Source: 950DT from Huawei's Huawei Connect 2025 keynote (Sept 18, 2025); 960DT as reported by Tom's Hardware from Huawei Connect 2026 (Sept 17, 2026).
Multiple = 960DT / 950DT (ARC Research calculation). Huawei's 2025 roadmap said the 960 generation would double interconnect to 4 TB/s; the 2026 report gives 2.2 TB/s.

SO WHAT

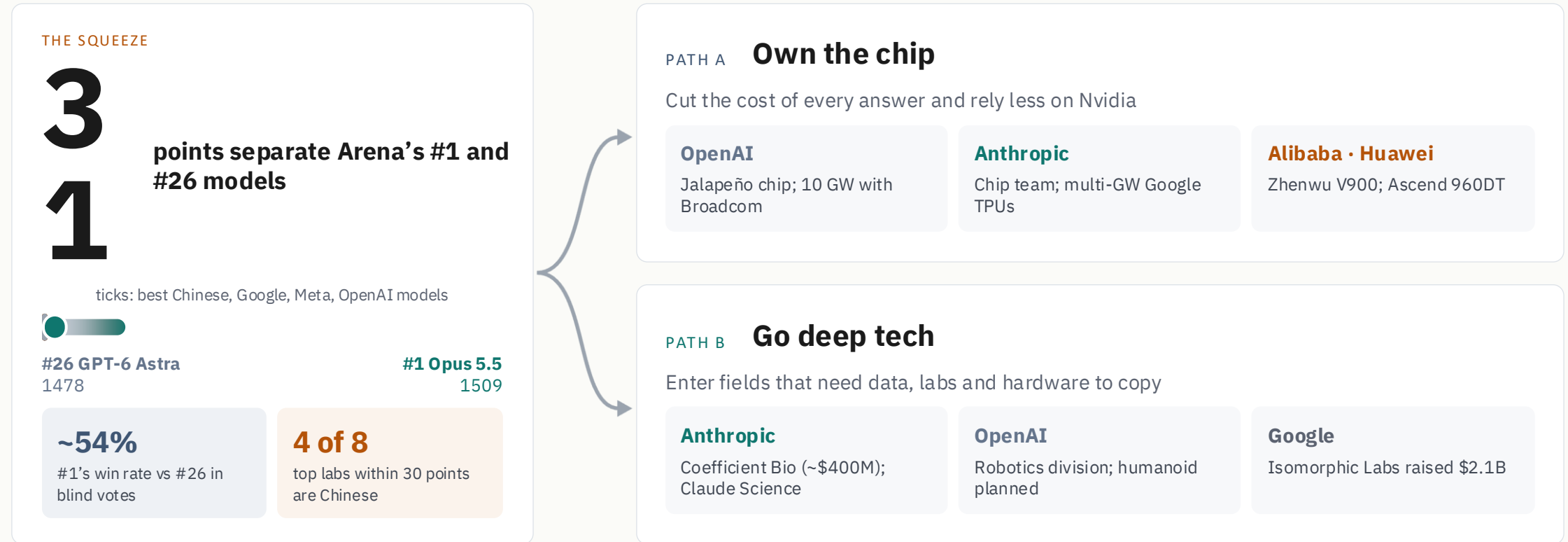
950DT (2026): Huawei's current chip for training and for generating answers

960DT: 2× compute and memory, 2.4× bandwidth, due Q1 2027

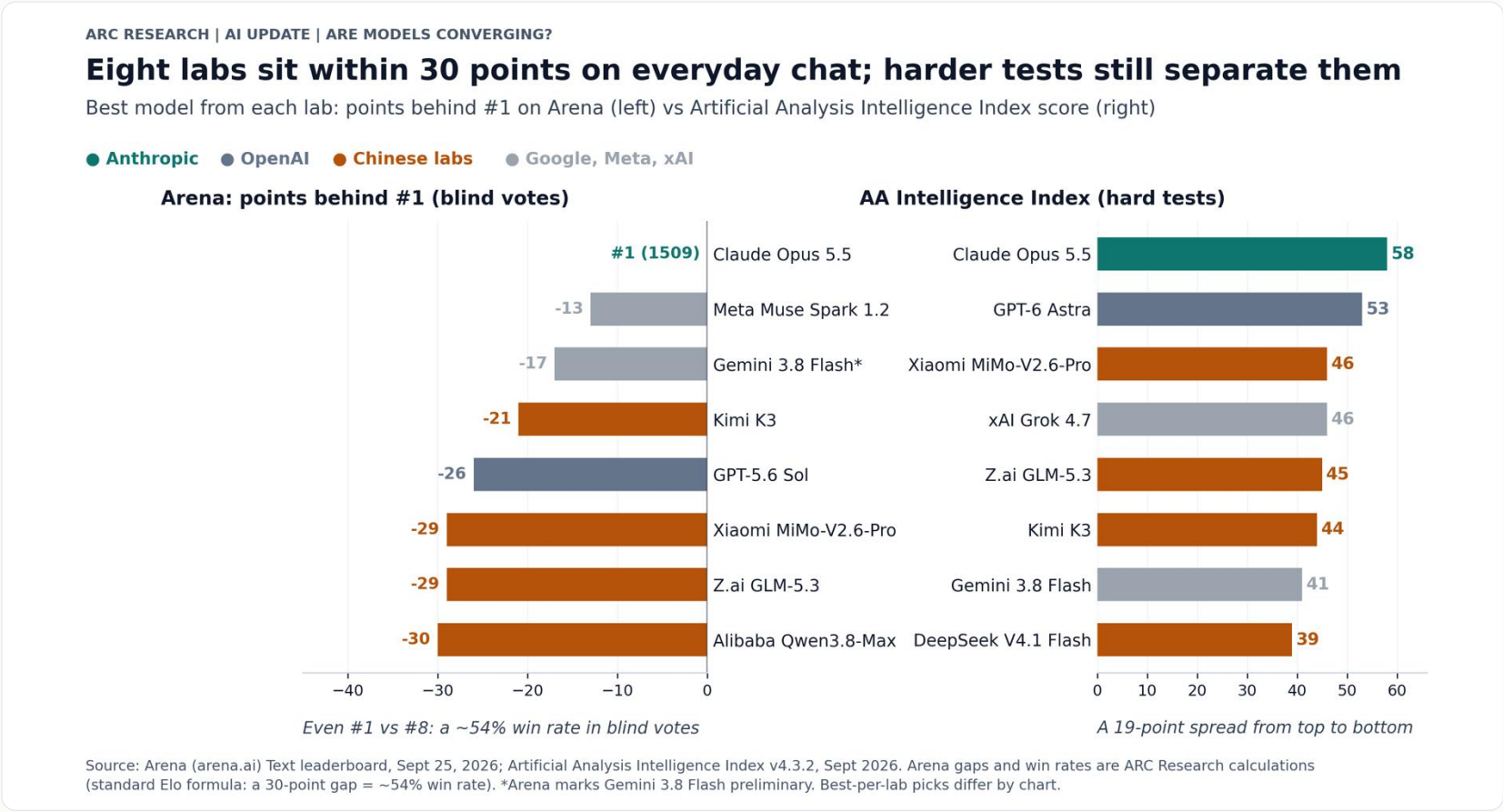
Still about half the FP8 compute of Nvidia's B300, per TheNextWeb



Models converge; labs turn to chips and deep tech



Close on everyday chat, still apart on hard tests



SO WHAT

For everyday answers, 8 labs are close to interchangeable

Four of those eight are Chinese, most with open weights

The frontier lead now lives on hard, long agentic tasks

Own the chip: the big labs now back custom silicon

WHO	CHIP	BUILT WITH	STATUS	GOAL
OpenAI	Jalapeño inference chip	Broadcom	Unveiled Jun 24; part of a 10 GW rollout, H2 2026–2029	Cheaper answers, less reliance on Nvidia
Anthropic	In-house chip team + Google TPUs	Google, Broadcom	Team confirmed Aug 6; multi-GW of TPUs from 2027; no chip date	Cut “the cost of every query”
Alibaba	Zhenwu V900	T-Head (in-house)	Unveiled Sept 22; mass production Q1 2027	Train 5–10T-parameter Qwen models
Huawei	Ascend 960DT	In-house	Pulled forward to Q1 2027	China’s main Nvidia alternative; DeepSeek bought 160K 950DTs

~119K

output tokens Opus 5.5 uses per hard task: agents burn tokens, so serving costs add up fast

Nvidia

the default supplier today; custom chips reduce that dependence

Export controls

block China from Nvidia’s top chips, forcing Alibaba and Huawei to build their own

Go deep tech: biology and robots are harder to copy

BIOLOGY & DRUG DISCOVERY		ROBOTICS	
Anthropic leads the push		OpenAI enters a China-led market	
OCT 2025	Anthropic launches Claude for Life Sciences (Benchling, 10x Genomics)	MAY 31	OpenAI launches a robotics division; 19 open roles by Sept
APR 2026	Anthropic buys drug-discovery startup Coefficient Bio (~\$400M)	SEP 3	OpenAI Altman: “We will definitely do a humanoid”
MAY 2026	Google Isomorphic Labs raises \$2.1B to design drugs with AI	H1 2026	Market ~19,100 humanoids shipped (+272%); 97%+ from Chinese makers
JUN 2026	Anthropic launches Claude Science, now a flagship product	—	Anthropic no robotics effort found

WHY IT’S A MOAT	Private lab data	Wet labs & trials	Regulators	Hardware & factories
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Reality check: pharma pays well, but drugs and robots take years to pay off; coding is still where the revenue is today.

Six terms behind this week's news

Term	What it means
MoE	Many expert sub-networks; only a few run per token
Cost per task	Price × tokens used to finish one task; fairer than list price
Intelligence Index	Artificial Analysis's independent score across many evaluations
FP8 / FP4	8- and 4-bit number formats: less memory, faster math
Inter-chip bandwidth	How fast accelerators share data when a model spans many chips
Open weights	Downloadable model parameters; license terms still vary

Sources

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- [The New Stack: GPT-6 Sol and Luna](#)
- [VentureBeat: Claude Opus 5.5](#)
- [Quartz: Claude Opus 5.5 pricing](#)
- [AlphaCorp: Opus 5.5 benchmarks](#)
- [Artificial Analysis: Claude Opus 5.5](#)
- [Artificial Analysis: GPT-6 Sol and Luna](#)
- [OfficeChai: AA Intelligence Index scores](#)
- [VentureBeat: Xiaomi MiMo-V2.6](#)
- [TechNode: MiMo-V2.6 open-sourced](#)
- [Artificial Analysis: MiMo-V2.6-Pro](#)
- [AI/TLDR: MiMo-V2.6 live RL dashboard](#)
- [TeknoGadyet: Qwen and HONOR partnership](#)
- [Huawei: Connect 2025 keynote \(Ascend 950\)](#)
- [Tom's Hardware: Ascend 960 roadmap](#)
- [TheNextWeb: Huawei Connect 2026](#)
- [Bloomberg \(via Yahoo\): 950DT price rise](#)
- [Arena: Text leaderboard \(Sept 25\)](#)
- [MindStudio: Opus 5 reception](#)
- [Gizmodo: Businesses shun Fable 5](#)
- [Ramp AI Index, Sept 2026](#)
- [TechCrunch: Anthropic buys Coefficient Bio](#)
- [Hugging Face: MiMo-V2.6-Pro-RL model card](#)
- [TechCrunch: OpenAI Jalapeño chip](#)
- [Forbes: Anthropic chip team](#)
- [OpenAI: Broadcom 10 GW collaboration](#)
- [CNBC: Claude for Life Sciences](#)
- [Arena: Text Pareto frontier](#)
- [The Decoder: Opus 5.5 and “Claudish”](#)
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- [Alibaba Cloud Apsara release \(via VIR\)](#)
- [Pandaily: Qwen 4 in training](#)
- [Alizila: Qwen Intelligence launch](#)
- [Gadget Pilipinas: Qwen Intelligence and HONOR](#)
- [CellCog: Qwen 4 status](#)
- [TechNode: Zhenwu V900](#)
- [Technology.org: Zhenwu V900](#)
- [Tom's Hardware: Zhenwu V900 and Ascend 960PR](#)
- [Yahoo Finance: Huawei Ascend 960](#)
- [The AI Insider: Ascend 960DT](#)
- [OpenCode: Kimi K4 listing](#)
- [DeepSeek: V4.1-Flash note \(V4.1-Pro\)](#)
- [Polymarket: Muse Spark 1.4 release](#)
- [Manifold: Kimi K4 in 2026](#)
- [CellCog: GLM release cadence](#)
- [The Standard: DeepSeek and Huawei chips](#)
- [TechNode: DeepSeek 160,000 Ascend chips](#)
- [The Register: R2 training on Ascend \(FT\)](#)
- [PrismML: Bonsai 2 27B launch](#)
- [MarkTechPost: Ternary Bonsai 2 27B](#)
- [MIT Tech Review: Claude Science](#)
- [Forbes: OpenAI humanoid robot](#)
- [Bloomberg: Isomorphic Labs \\$2.1B](#)

